

Exercise 5:

```
a = 0; % lower limit
b=10; % upper limit
y = [0 1 3 5 7 8 9 10]; % distance from the bank
H = [0 1 1.5 3 3.5 3.2 2 0]; % depth
U=[0 0.1 0.12 0.2 0.25 0.3 0.15 0]; % velocity
Ac_numerical = tabular_trapezoidal(y,H);
H_bar = Ac_numerical/b; % average depth
Q_FlowRate = tabular_trapezoidal(y,H.*U);
U_bar = Q_FlowRate/Ac_numerical; % average velocity
fprintf("Cross sectional area Ac = %f \n" + ...
        "Average depth H_bar = %f \n" + ...
        "Flow rate Q = %f \n" + ...
        "Average velocity U_bar = %f \n",Ac_numerical,H_bar,Q_FlowRate,U_bar)
```

```
Cross sectional area Ac = 20.950000
Average depth H_bar = 2.095000
Flow rate Q = 4.282500
Average velocity U_bar = 0.204415
```

```
function [I] = tabular_trapezoidal(x,y) % x and y vectors
if (length(x)==length(y))
    I = 0;
    n = length(x);
    for i=1:n-1
        h = x(i+1)-x(i); % width of segment i
        I = I + h*(y(i+1)+y(i))/2;
    end
else
    disp("Vectors x and y must be same length.")
end
end

function [y]=linear_interpolation(X,Y,xq)
y = interp1(X,Y,xq);
end
```